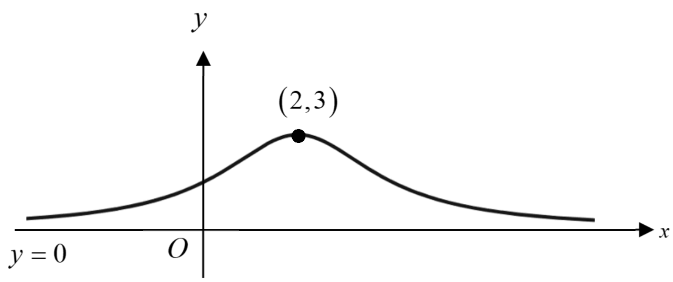
2021 JC1 H2 Math Promo Solution

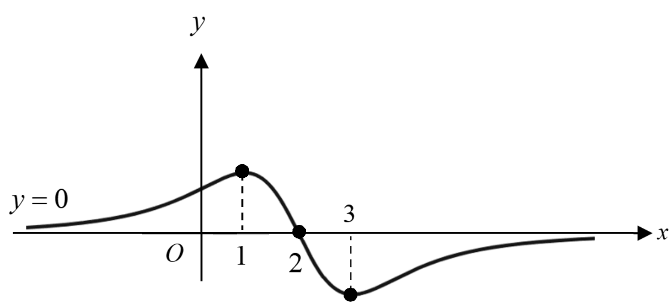
1.



The diagram shows the curve $y=\mathrm{f}(x)$. The curve has an asymptote $y=0$ and a maximum point at $(2,3)$. It is given that f is concave downwards for $1 \le x \le 3$.

Sketch the graph of $y=\mathrm{f}'(x)$, stating the equations of any asymptotes, the $x-$coordinates of any stationary points and any points of intersection with the $x-$axis. [3]

Solution



2. In a convergent geometric series that only consists of positive terms, the sum of the first four terms is 80 times the sum of all the remaining terms. Find the common ratio. [4]

<u>Solution</u>

Let a be the first term of the infinite geometric progression (GP),
and r be the common ratio.

Sum of first four terms = 80 × sum of all the remaining terms

$$S_4 = 80\left[S_\infty - S_4\right]$$

$$\therefore\ \frac{a(1-r^4)}{1-r} = 80\frac{ar^4}{1-r}$$

$$1-r^4 = 80r^4 \quad (\because a \neq 0)$$

$$1 = 81r^4$$

$$r^4 = \frac{1}{81}$$

$$r = \pm\frac{1}{3}$$

$\because$ The terms of the geometric series are all positive,
common ratio r is $\frac{1}{3}$.

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3. Differentiate the following expressions with respect to x, leaving your answers in terms of x,

(a) $\cos^{-1}\sqrt{x^3}$, giving your answer in non-trigonometric form, [2]

(b) $\operatorname{cosec}\left(\ln\left(x^2+3\right)\right)$, [2]

(c) $\left(x^2+1\right)^x$. [3]

Solution

(a)

Method 1 (Chain Rule)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}\sqrt{x^3}\right)=\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}\left(x^{\frac{3}{2}}\right)\right)$$

$$=-\frac{1}{\sqrt{1-\left(x^{\frac{3}{2}}\right)^2}}\left(\frac{3}{2}x^{\frac{1}{2}}\right)$$

$$=-\frac{3\sqrt{x}}{2\sqrt{1-x^3}}$$

Method 2 (Chain Rule)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}\sqrt{x^3}\right)=\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}\left(x^3\right)^{\frac{1}{2}}\right)$$

$$=-\frac{1}{\sqrt{1-\left(x^3\right)}}\left(\frac{1}{2}\frac{1}{\sqrt{x^3}}\right)\left(3x^2\right)$$

$$=-\frac{3x^2}{2\sqrt{x^3}\sqrt{1-x^3}}$$

$$=-\frac{3\sqrt{x}}{2\sqrt{1-x^3}}$$

Method 3 (Implicit Diff)

Let $\quad y=\cos^{-1}\sqrt{x^3}$

$$\cos y=x^{\frac{3}{2}}$$

Differentiating w.r.t. x,

$$-\sin y\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{3}{2}\sqrt{x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{3\sqrt{x}}{2\sin y}$$

$$=-\frac{3\sqrt{x}}{2\sqrt{\sin^2 y}}$$

$$= -\frac{3\sqrt{x}}{2\sqrt{1-\cos^2 y}} = -\frac{3\sqrt{x}}{2\sqrt{1-x^3}}$$ [Final answer in terms of x]

(b)

Method 1 (Chain rule, formula in MF26)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\operatorname{cosec}\left(\ln(x^2+3)\right)\right]$$

$$= -\operatorname{cosec}\left(\ln(x^2+3)\right)\cot\left(\ln(x^2+3)\right)\cdot\left(\frac{1}{x^2+3}(2x)\right)$$

$$= -\frac{2x\operatorname{cosec}\left(\ln(x^2+3)\right)\cot\left(\ln(x^2+3)\right)}{x^2+3}$$

Method 2 (Chain rule)

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\operatorname{cosec}\left(\ln(x^2+3)\right)\right]$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{\sin\left(\ln(x^2+3)\right)}\right]$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\left[\sin\left(\ln(x^2+3)\right)\right]^{-1}$$

$$= -\left[\sin\left(\ln(x^2+3)\right)\right]^{-2}\cdot\left[\cos\left(\ln(x^2+3)\right)\right]\left(\frac{1}{x^2+3}(2x)\right)$$

$$= -\frac{2x\ \cos\left(\ln(x^2+3)\right)}{(x^2+3)\ \sin^2\left(\ln(x^2+3)\right)}$$

(c)

Method 1: [Implicit Diff]

Let $y=(x^2+1)^x$

$\ln y = \ln(x^2+1)^x$

$\ln y = x\ln(x^2+1)$

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = x\left(\frac{2x}{x^2+1}\right)+(1)\ln(x^2+1)$$ [Product Rule]

$$\frac{\mathrm{d}y}{\mathrm{d}x} = y\left[\frac{2x^2}{x^2+1}+\ln(x^2+1)\right]$$

$$= (x^2+1)^x\left[\frac{2x^2}{x^2+1}+\ln(x^2+1)\right]$$ [Final answer in terms of x]

Method 2 (Chain rule)

Let $y = (x^2+1)^x$

$$= \mathrm{e}^{\ln(x^2+1)^x}$$

$$= \mathrm{e}^{x\ln(x^2+1)}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{x\ln(x^2+1)}\left[(1)\ln(x^2+1) + x\left(\frac{2x}{x^2+1}\right)\right]$$ [Product rule]

$$= \mathrm{e}^{x\ln(x^2+1)}\left[\ln(x^2+1) + \frac{2x^2}{x^2+1}\right]$$

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4. The points $A(1,0,-2)$, $B(3,-1,-2)$ and $C(-3,7,0)$ lie on plane p_1.

Another plane p_2 has equation $3x-y+2z=3$.

(i) Find a vector equation of plane p_1 in the form $\mathbf{r}\cdot\mathbf{n}=d$. [3]

(ii) Find the acute angle between p_1 and p_2. [2]

The equation of plane p_3 is given to be $-9x+3y-6z=7$.

(iii) Find the shortest distance between p_2 and p_3. [3]

<u>Solution</u>

(i)

normal to plane p_1

$$=\overrightarrow{AB}\times\overrightarrow{AC}=\begin{pmatrix}3-1\\-1-0\\-2-(-2)\end{pmatrix}\times\begin{pmatrix}-3-1\\7-0\\0-(-2)\end{pmatrix}=\begin{pmatrix}2\\-1\\0\end{pmatrix}\times\begin{pmatrix}-4\\7\\2\end{pmatrix}=\begin{pmatrix}-2\\-4\\10\end{pmatrix}=-2\begin{pmatrix}1\\2\\-5\end{pmatrix}$$

OR

normal to plane p_1

$$=\overrightarrow{AB}\times\overrightarrow{BC}=\begin{pmatrix}3-1\\-1-0\\-2-(-2)\end{pmatrix}\times\begin{pmatrix}-3-3\\7-(-1)\\0-(-2)\end{pmatrix}=\begin{pmatrix}2\\-1\\0\end{pmatrix}\times\begin{pmatrix}-6\\8\\2\end{pmatrix}=\begin{pmatrix}-2\\-4\\10\end{pmatrix}=-2\begin{pmatrix}1\\2\\-5\end{pmatrix}$$

Equation of plane p_1:

$$\underset{\sim}{r}\cdot\begin{pmatrix}1\\2\\-5\end{pmatrix}=\begin{pmatrix}1\\0\\-2\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-5\end{pmatrix}=11$$

$$\boxed{\text{Or } \underset{\sim}{r}\cdot\begin{pmatrix}-2\\-4\\10\end{pmatrix}=\begin{pmatrix}1\\0\\-2\end{pmatrix}\cdot\begin{pmatrix}-2\\-4\\10\end{pmatrix}=-22\ \Rightarrow \underset{\sim}{r}\cdot\begin{pmatrix}1\\2\\-5\end{pmatrix}=11}$$

(ii)

Equation of plane p_2: $\underset{\sim}{r}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=3$

acute angle between p_1 and p_2

$= \cos^{-1} \dfrac{\left| \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \right|}{\sqrt{30}\sqrt{14}}$

$= \cos^{-1} \dfrac{9}{\sqrt{30}\sqrt{14}}$

$= 1.12 \quad \text{OR} \quad 64.0^{\circ}$

OR

$= \cos^{-1} \dfrac{\left| \begin{pmatrix} -2 \\ -4 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \right|}{\sqrt{120}\sqrt{14}}$

$= \cos^{-1} \dfrac{18}{\sqrt{120}\sqrt{14}}$

$= 1.12 \quad \text{OR} \quad 64.0^{\circ}$

(iii)

Method 1 (Using length of projection)

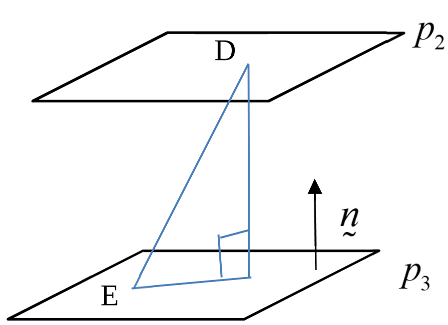
$p_2 : \underset{\sim}{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 3$ and $p_3 : -9x + 3y - 6z = 7 \Rightarrow \underset{\sim}{r} \cdot \begin{pmatrix} -9 \\ 3 \\ -6 \end{pmatrix} = 7$

So the two planes are parallel.

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 3$ and $\begin{pmatrix} 0 \\ 0 \\ -\frac{7}{6} \end{pmatrix} \cdot \begin{pmatrix} -9 \\ 3 \\ -6 \end{pmatrix} = 7$

Select points $D(1,0,0)$ and $E\left(0,0,-\frac{7}{6}\right)$ on planes p_2 and p_3 respectively.

Shortest distance between p_2 and p_3



$= \dfrac{\left| \overline{DE} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \right|}{\sqrt{14}}$

$= \dfrac{\left| \begin{pmatrix} 0-1 \\ 0-0 \\ -\frac{7}{6}-0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \right|}{\sqrt{14}}$

$= \dfrac{\left| \begin{pmatrix} -1 \\ 0 \\ -\frac{7}{6} \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \right|}{\sqrt{14}}$

$= \dfrac{16}{3\sqrt{14}}$

$= 1.43$ units

Method 2: [Using perpendicular distance of plane from the origin]

$$p_2:\ \underset{\sim}{r}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=3 \Rightarrow \underset{\sim}{r}\cdot\frac{1}{\sqrt{14}}\begin{pmatrix}3\\-1\\2\end{pmatrix}=\frac{3}{\sqrt{14}}$$

$$p_3:\ \underset{\sim}{r}\cdot\begin{pmatrix}-9\\3\\-6\end{pmatrix}=7 \Rightarrow \underset{\sim}{r}\cdot\frac{1}{3\sqrt{14}}\begin{pmatrix}3\\-1\\2\end{pmatrix}=-\frac{7}{3\sqrt{14}}$$ [match the direction of the normal]

Hence, perpendicular distance = $\dfrac{3}{\sqrt{14}}-\left(-\dfrac{7}{3\sqrt{14}}\right)=\dfrac{9+7}{3\sqrt{14}}=\dfrac{16}{3\sqrt{14}}$

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5. **(a)** **(i)** Express $\dfrac{2r^2+1}{r^2-1}$ in the form $A+\dfrac{B}{r-1}+\dfrac{C}{r+1}$, where A, B and C are constants to be found. [2]

(ii) Hence find $\displaystyle\sum_{r=2}^{n}\frac{2r^2+1}{r^2-1}$ in terms of n. [4]

(b) Express $\displaystyle\sum_{r=1}^{2n}\left[(-1)^{r+1}2^r\right]$ in the form $\dfrac{p}{q}\left[s\left(4^n\right)+t\right]$, where p, q ,s and t are integers to be determined. [3]

Solution

(ai)

$$\frac{2r^2+1}{r^2-1}=A+\frac{B}{r-1}+\frac{C}{r+1}$$

$$2r^2+1=A(r-1)(r+1)+B(r+1)+C(r-1)$$

Comparing coefficients, $2=A$

$$0=B+C$$

$$1=-A+B-C$$

Solving, $A=2, B=\dfrac{3}{2}$ and $C=-\dfrac{3}{2}$

Hence, $\dfrac{2r^2+1}{r^2-1}=2+\dfrac{3}{2(r-1)}-\dfrac{3}{2(r+1)}$

(aii)

$$\sum_{r=2}^{n}\frac{2r^2+1}{r^2-1}=\sum_{r=2}^{n}\left[2+\frac{3}{2(r-1)}-\frac{3}{2(r+1)}\right]$$

$$=\sum_{r=2}^{n}2+\frac{3}{2}\sum_{r=2}^{n}\left[\frac{1}{(r-1)}-\frac{1}{(r+1)}\right]$$

$$=(n-2+1)(2)+\frac{3}{2}\left[\begin{array}{l}\frac{1}{1}-\frac{1}{3}+\\ \frac{1}{2}-\frac{1}{4}+\\ \frac{1}{3}-\frac{1}{5}+\\ \frac{1}{4}-\frac{1}{6}+\\ \quad\vdots\\ \frac{1}{n-3}-\frac{1}{n-1}+\\ \frac{1}{n-2}-\frac{1}{n}+\\ \frac{1}{n-1}-\frac{1}{n+1}\end{array}\right]$$

$$= 2(n-1) + \frac{3}{2}\left[1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}\right]$$

$$= 2(n-1) + \frac{3}{2}\left[\frac{3}{2} - \frac{1}{n} - \frac{1}{n+1}\right]$$

(b)

[Should list out to see the pattern]

Method 1

$$\sum_{r=1}^{2n}\left[(-1)^{r+1}2^r\right] = 2^1 - 2^2 + 2^3 - 2^4 + \ldots + 2^{2n-1} - 2^{2n}$$

[G.P. with first term = 2, common ratio = -2, nbr of terms is $2n$]

$$= \frac{(2)\left((-2)^{2n} - 1\right)}{(-2) - 1}$$

$$= \frac{2}{3}\left(1 - 4^n\right)$$

Method 2

$$\sum_{r=1}^{2n}\left[(-1)^{r+1}2^r\right] = 2^1 - 2^2 + 2^3 - 2^4 + \ldots + 2^{2n-1} - 2^{2n}$$

$$= \left(2^1 + 2^3 + \ldots + 2^{2n-1}\right) - \left(2^2 + 2^4 + \ldots 2^{2n}\right)$$

$$= \frac{2\left(4^n - 1\right)}{4-1} - \frac{2^2\left(4^n - 1\right)}{4-1}$$

$$= \frac{2}{3}\left(4^n - 1 - 2\left(4^n\right) + 2\right)$$

$$= \frac{2}{3}\left(1 - 4^n\right)$$

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6. The curve C is defined by the parametric equations

$$x = \theta + \frac{1}{2}\sin 2\theta,\ y = 2\tan\theta \qquad \text{where } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

(i) Show that $\frac{dy}{dx} = \frac{1}{\cos^k\theta}$, where k is an integer to be determined. [3]

The lines T and N are the tangent and normal to C at the point where $\theta = \frac{\pi}{4}$ respectively.

(ii) Find the equations of T and N, leaving your answers in exact form. [3]

(iii) Find the area enclosed by T, N and the y-axis. [2]

Solution

(i)

$$\frac{dx}{d\theta} = 1 + \frac{1}{2}(2)\cos 2\theta,\ \frac{dy}{d\theta} = 2\sec^2\theta$$

$$\frac{dy}{dx} = \frac{2\sec^2\theta}{1+\cos 2\theta} = \frac{2\sec^2\theta}{2\cos^2\theta} = \frac{1}{\cos^4\theta}$$

So, $k = 4$

(ii)

When $\theta = \frac{\pi}{4}$, $x = \frac{\pi}{4} + \frac{1}{2}$, $y = 2$, $\frac{dy}{dx} = 4$

Equation of tangent : $y - 2 = 4\left[x - \left(\frac{\pi}{4} + \frac{1}{2}\right)\right]$

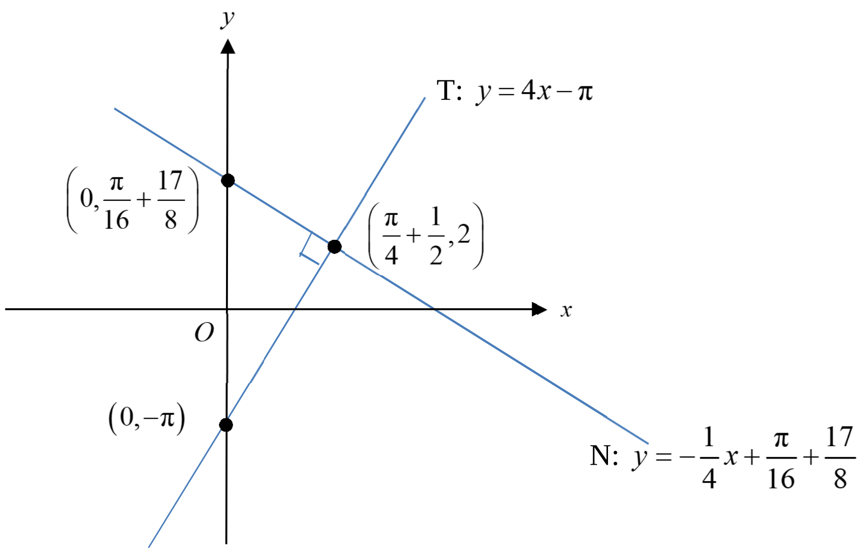
$y = 4x - \pi$

Equation of normal : $y - 2 = -\frac{1}{4}\left[x - \left(\frac{\pi}{4} + \frac{1}{2}\right)\right]$

$y = -\frac{1}{4}x + \frac{\pi}{16} + \frac{17}{8}$

(iii) [Note: last part can use GC since non-exact answer allowed.]

[Note: no need to manual find point of intersection of T and N since occurs at $\theta = \frac{\pi}{4}$]



Coordinates of y-intercepts are $\left(0, \frac{\pi}{16} + \frac{17}{8}\right)$ and $(0, -\pi)$

Area enclosed $= \frac{1}{2}bh = \frac{1}{2}\left(\frac{\pi}{16} + \frac{17}{8} + \pi\right)\left(\frac{\pi}{4} + \frac{1}{2}\right)$ =3.51

Alternative Method

Area enclosed $= \frac{1}{2}\begin{vmatrix} 0 & \frac{\pi}{4} + \frac{1}{2} & 0 & 0 \\ -\pi & 2 & \frac{\pi}{16} + \frac{17}{8} & -\pi \end{vmatrix}$

$= \frac{1}{2}\left|\left(\frac{\pi}{4} + \frac{1}{2}\right)\left(\frac{\pi}{16} + \frac{17}{8}\right) - (-\pi)\left(\frac{\pi}{4} + \frac{1}{2}\right)\right|$

$= 3.51$ square units

7. It is given that $y=\sin\left[\ln\left(1+2x\right)\right]$.

(i) Show that $\left(1+2x\right)^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2\left(1+2x\right)\frac{\mathrm{d}y}{\mathrm{d}x}+ky=0$, where k is a constant to be found.

Hence, find the first three non-zero terms of the Maclaurin expansion for y. [6]

(ii) Using standard series from the List of Formulae (MF26), verify the correctness of your result from part **(i)** up to and including the term in x^3.

Explain why the expansion is not valid when $x=-\frac{1}{2}$. [3]

<u>Solution</u>

(i)

$y=\sin\left[\ln\left(1+2x\right)\right]$

Differentiating with respect to x,

$\frac{\mathrm{d}y}{\mathrm{d}x}=\cos\left[\ln\left(1+2x\right)\right].\frac{2}{1+2x}$ [Chain rule]

$\left(1+2x\right)\frac{\mathrm{d}y}{\mathrm{d}x}=2\cos\left[\ln\left(1+2x\right)\right]$

Differentiating with respect to x,

$\left(1+2x\right)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2\frac{\mathrm{d}y}{\mathrm{d}x}=-2\sin\left[\ln\left(1+2x\right)\right].\frac{2}{1+2x}$ [Product rule]

$\left(1+2x\right)^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2\left(1+2x\right)\frac{\mathrm{d}y}{\mathrm{d}x}+4y=0$ (Shown)

So $k=4$

Differentiating with respect to x,

$\left(1+2x\right)^2\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+4\left(1+2x\right)^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2\left(1+2x\right)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+4\frac{\mathrm{d}y}{\mathrm{d}x}+4\frac{\mathrm{d}y}{\mathrm{d}x}=0$

$\left(1+2x\right)^2\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+6\left(1+2x\right)^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+8\frac{\mathrm{d}y}{\mathrm{d}x}=0$

When $x=0$,

$y=0$

$\frac{\mathrm{d}y}{\mathrm{d}x}=2$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+2(2)=0\Rightarrow\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=-4$

$\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+6(-4)+8(2)=0\Rightarrow\frac{\mathrm{d}^3y}{\mathrm{d}x^3}=8$

$\therefore y=0+2x-\frac{4}{2!}x^2+\frac{8}{3!}x^3+\ldots$

$=2x-2x^2+\frac{4}{3}x^3+\ldots$ (up to x^3)

(ii)

Using standard series from MF26,

$y = \sin\left[\ln\left(1+2x\right)\right]$

$= \sin\left[2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} + ...\right]$

$= \sin\left[2x - 2x^2 + \frac{8}{3}x^3 + ...\right]$

$= \left[2x - 2x^2 + \frac{8}{3}x^3 + ...\right] - \frac{1}{3!}\left[2x - 2x^2 + \frac{8}{3}x^3 + ...\right]^3 + ...$

$= 2x - 2x^2 + \frac{8}{3}x^3 + ... - \frac{1}{6}(2x)^3 + ...$

$= 2x - 2x^2 + \frac{8}{3}x^3 + ... - \frac{4}{3}x^3 + ...$

$= 2x - 2x^2 + \frac{4}{3}x^3 + ...$

Hence correct up to x^3 term.

From MF26

$\ln\left(1+x\right) = x - \frac{x^2}{2} + \frac{x^3}{3} ... \quad -1 < x \leq 1$

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - ...$

When $x = -\frac{1}{2}$, $\ln\left(1+2x\right)$ is undefined.

Or,

Since expansion for $\ln\left(1+2x\right)$ is valid only for $-1 < 2x \leq 1 \Rightarrow -\frac{1}{2} < x \leq \frac{1}{2}$

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8 **(i)** The arithmetic progression is grouped into sets of integers, such that the n^{th} set contains n integers as shown.

$$\{\ 1\ \},\{\ 3,\ 5\ \},\{\ 7,\ 9,\ 11\ \},\{\ 13,\ 15,\ 17,\ 19\ \},\ldots\ \ldots$$

(a) Find the total number of terms in the first n sets, and hence show that the last term of the n^{th} set is n^2+n-1. [2]

(b) Find the first term of the n^{th} set. [2]

(c) Show that the sum of the terms in the n^{th} set is n^3. [1]

(ii) Hence, prove that $1^3+2^3+3^3+\ldots\ldots+n^3=\dfrac{1}{4}n^2\left(n+1\right)^2$. [2]

Solution

(i)(a)

[Identify the pattern]

$$\underbrace{\{\ 1\ \}}_{\substack{1^{\text{st}}\ \text{set}\\(1\ \text{term})}},\underbrace{\{\ 3,\ 5\ \}}_{\substack{2^{\text{nd}}\ \text{set}\\(2\ \text{terms})}},\underbrace{\{\ 7,\ 9,\ 11\ \}}_{\substack{3^{\text{rd}}\ \text{set}\\(3\ \text{terms})}},\underbrace{\{\ 13,\ 15,\ 17,\ 19\ \}}_{\substack{4^{\text{th}}\ \text{set}\\(4\ \text{terms})}},\ldots\ \ldots,\underbrace{\{\ \ldots\ \ldots\ \}}_{\substack{n^{\text{th}}\ \text{set}\\(n\ \text{terms})}}$$

Total number of terms in the first n sets

$=1+2+3+\ldots+n$

$=\dfrac{n}{2}(1+n)$

OR

Total number of terms in the first n sets

$=1+2+3+\ldots+n$

$=\dfrac{n}{2}\left(2(1)+(n-1)(1)\right)$

$=\dfrac{n}{2}(1+n)$

Last term of the n^{th} set

$=\left[\dfrac{n}{2}(1+n)\right]^{\text{th}}$ term of the A.P. with first term 1 and common difference 2.

$=1+\left[\dfrac{n}{2}(1+n)-1\right](2)$ [$T_n=a+\left(n-1\right)d$]

$=1+n(1+n)-2$

$=n^2+n-1$

Alternatively

Let u_k denote the k^{th} term of the AP.

$\Rightarrow u_k=1+(k-1)(2)$

$\qquad =2k-1$

Last term of the n^{th} set, $u_{\frac{n}{2}(1+n)}=2\left[\dfrac{n}{2}(1+n)\right]-1$

$=n(1+n)-1$

$=n^2+n-1$

(i)(b)

Method 1 (Shortest method)

Since from (ia) last term of nth set is n^2+n-1.
So, last term of the $(n-1)^{th}$ set

$=(n-1)^2+(n-1)-1$

$=n^2-2n+1+n-2$

$=n^2-n-1$

First term of the n^{th} set
= Last term of the $(n-1)^{th}$ set + 2

$=n^2-n+1$

Method 2

First term of the $(n+1)^{th}$ set
= Last term of the n^{th} set + 2

$=(n^2+n-1)+2$

$=n^2+n+1$

First term of the n^{th} set, i.e. $[(n-1)+1]^{th}$ set

$=(n-1)^2+(n-1)+1$

$=n^2-2n+1+n$

$=n^2-n+1$

Method 3

First term of the n^{th} set
= Last term of the n^{th} set − $(n-1)$(common difference of A.P.)

$=(n^2+n-1)-(n-1)(2)$

$=n^2-n+1$

Method 4

Let a denote the first term of the n^{th} set.

$$\underbrace{n^2+n-1}_{\substack{n^{\text{th}}\text{ (last) term}\\ \text{of the set}}}=a+(n-1)(2)$$

$\Rightarrow a=n^2+n-1-2(n-1)$

$\quad =n^2-n+1$

Method 5:

Total number of terms in the first $(n-1)$ sets

$=1+2+3+...+(n-1)$

$=\frac{n-1}{2}\left(1+(n-1)\right) \;=\frac{n(n-1)}{2}$

First term of the n^{th} set

$$= \left[\frac{n}{2}(n-1)+1\right]^{\text{th}} \text{ term of the arithmetic progression (AP),}$$

$$= 1 + \left[\frac{n}{2}(n-1)+1-1\right](2)$$

$$= 1 + n(n-1)$$

$$= n^2 - n + 1$$

Method 6: (Pattern observation – Guess and extend)

$$\underset{1^{\text{st}}\ \text{set}}{\{\overset{1^2=1}{1}\}},\ \underset{2^{\text{nd}}\ \text{set}}{\{\overset{-1}{3}, \overset{2^2=4}{5}\}},\ \underset{3^{\text{rd}}\ \text{set}}{\{\overset{-2}{7}, \overset{3^2=9}{9}, 11\}},\ \underset{4^{\text{th}}\ \text{set}}{\{\overset{-3}{13}, 15, \overset{4^2=16}{17}, 19\}},\ \dots\ \dots$$

Set	First term of set
1	$1 = 1^2 - 0$
2	$3 = 2^2 - 1$
3	$7 = 3^2 - 2$
4	$13 = 4^2 - 3$
…	…
n	$n^2 - (n-1)$

$\therefore$ Last term of n^{th} set $= n^2 - n + 1$

(i)(c)

Method 1:

Sum of the terms in the n^{th} set

$$= \underbrace{(n^2-n+1) + \ \dots\ \dots\ + (n^2+n-1)}_{\text{Terms of the } n^{th} \text{ set: } n \text{ terms in A.P.}}$$

$$= \frac{n}{2}\left[(n^2-n+1)+(n^2+n-1)\right]$$

$$= \frac{n}{2}\left(2n^2\right)$$

$$= n^3 \qquad \text{(Shown)}$$

Method 2:

Sum of the terms in the n^{th} set

$$= \underbrace{(n^2 - n + 1) + \ \dots\dots\ + (n^2 + n - 1)}_{\text{Terms of the } n^{th} \text{ set: } n \text{ terms in A.P.}}$$

$$= \frac{n}{2}\left[2(n^2 - n + 1) + (n-1)(2)\right]$$

$$= \frac{n}{2}\left(2n^2 \cancel{-2n} + \cancel{2} + \cancel{2n} \cancel{-2}\right)$$

$$= n^3 \qquad \text{(Shown)}$$

(ii)

Observe that from **(i)(c)** sum of terms in the nth bracket is n^3.

Hence sum of terms in 1st bracket is 1^3

sum of terms in 2^{nd} bracket is 2^3

sum of terms in 3^{rd} bracket is 3^3

…

sum of terms in nth bracket is n^3

$1^3 + 2^3 + \dots + n^3$ = sum of terms in the first n brackets

$$\left(\text{A.P. with first term 1, common difference 1, nbr of terms } \frac{n}{2}(1+n)\right)$$

$$= \frac{\left[\frac{n}{2}(1+n)\right]}{2}\left[1 + \underbrace{(n^2 + n - 1)}_{\text{last term of first } n \text{ sets}}\right]$$

$$= \frac{1}{4}n(n+1)\left[n^2 + n\right]$$

$$= \frac{1}{4}n^2(n+1)^2 \quad = \text{RHS} \quad \text{(Shown)}$$

<u>Alternative Method</u>

$$\dots = \frac{\left[\frac{n}{2}(1+n)\right]}{2}\left[2(1) + \left[\frac{n}{2}(1+n) - 1\right](2)\right]$$

$$= \frac{1}{4}n(n+1)\left[n(1+n)\right]$$

$$= \frac{1}{4}n^2(n+1)^2 \quad = \text{RHS} \quad \text{(shown)}$$

---

9. With reference to the origin O, the points A and B have position vectors $\mathbf{a} = -\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = 2\mathbf{j} + 5\mathbf{k}$ respectively.

(i) Find a vector equation of the line l_1 that passes through point A and is parallel to the vector **a**. [1]

(ii) Find the exact length of projection of **b** on l_1. Hence find d, the exact perpendicular distance from the point B to l_1. [4]

(iii) Using the value of d found in part **(ii)**, find the position vector of the point C, the foot of perpendicular from the point B to l_1. [3]

(iv) The line l_2 passes through point B and is parallel to vector **b**. Find a cartesian equation of l_3 which is the reflection of l_2 in l_1. [3]

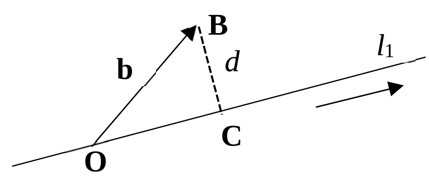
<u>Solution</u>

(i)

$$l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + k\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \lambda\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \ \lambda \in \mathbb{R}$$

or $l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + k\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \ k \in \mathbb{R}$

(ii)



length of projection of **b** on l_1 = $\left| \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} \cdot \frac{\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}}{\sqrt{(-1)^2 + (1)^2 + (2)^2}} \right|$

$$= \frac{12}{\sqrt{6}} = 2\sqrt{6}$$

$|\mathbf{b}| = \left| \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} \right| = \sqrt{29}$

By Pythagoras' Theorem,

d, Perpendicular distance from B to l_1 = $\sqrt{|\mathrm{b}|^2 - \left(2\sqrt{6}\right)^2}$

$$= \sqrt{29 - 4(6)} = \sqrt{5}$$

<u>Method 2</u>

Let C be foot of perpendicular from B to l_1, using length of projection of **b** on l_1 and O lies on l_1,

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB}$$

$$= 2\sqrt{6}\,\underset{\sim}{\hat{a}} - \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$$

$$= 2\sqrt{6} \times \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

$$\left|\overrightarrow{BC}\right| = \left|\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}\right| = \sqrt{5}$$

(iii)

Since C lies on l_1,

$$\mathbf{c} = \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \text{ for some } \lambda \in \mathbb{R}$$

$$\overrightarrow{BC} = \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} -\lambda \\ \lambda - 2 \\ 2\lambda - 5 \end{pmatrix}$$

$$\left|\overrightarrow{BC}\right| = \left|\begin{pmatrix} -\lambda \\ \lambda - 2 \\ 2\lambda - 5 \end{pmatrix}\right| = \sqrt{5}$$

$$\sqrt{(-\lambda)^2 + (\lambda - 2)^2 + (2\lambda - 5)^2} = \sqrt{5}$$

$$6\lambda^2 - 24\lambda + 24 = 0$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$(\lambda - 2)^2 = 0$$

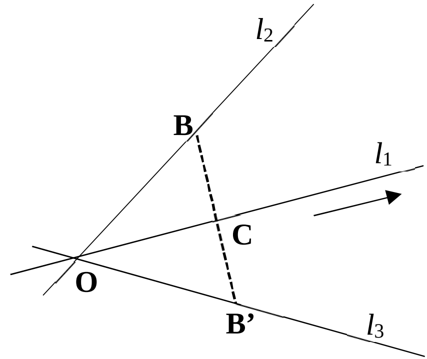
$$\lambda = 2$$

$$\mathbf{c} = 2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 4 \end{pmatrix}$$

(vi)

$$l_2 : \mathbf{r} = \mu \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix},\ \mu \in \mathbb{R}$$

Let B' be the point of reflection of B about the line l_1.
By mid-point theorem,



$$\overrightarrow{OC} = \frac{\overrightarrow{OB} + \overrightarrow{OB'}}{2}$$

$$\begin{pmatrix} -2 \\ 2 \\ 4 \end{pmatrix} = \frac{1}{2}\left[\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} + \overrightarrow{OB'}\right]$$

$$\overrightarrow{OB'} = 2\begin{pmatrix} -2 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \\ 3 \end{pmatrix}$$

Since l_3 parallel to $\overrightarrow{OB'}$ and the origin O lies on l_3,

$$l_3 : \mathbf{r} = \gamma \begin{pmatrix} -4 \\ 2 \\ 3 \end{pmatrix},\ \gamma \in \mathbb{R}$$

So Cartesian equation of l_3 is : $-\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$

or $\frac{-x-4}{4} = \frac{y-2}{2} = \frac{z-3}{3}$

10. The curves C_1 and C_2 have equations $y=\dfrac{2x^2+9}{x^2-4}$ and $\dfrac{x^2}{9}+\dfrac{y^2}{25}=1$ respectively.

(i) Find the equations of the asymptotes of the curve C_1. [3]

(ii) Sketch C_1 and C_2 on the same diagram, stating the equations of any asymptotes, coordinates of any points where C_1 or C_2 crosses the axes and any turning points. [5]

(iii) Find the x-coordinates of the points where the two curves intersect. [2]

(iv) Hence solve the inequality $-5\sqrt{1-\dfrac{1}{9}x^2}\le\dfrac{2x^2+9}{x^2-4}$. [2]

Solution

(i)

$$y=\frac{2x^2+9}{x^2-4}=2+\frac{17}{(x+2)(x-2)}$$

V.A. $x=-2,\ x=2$

and H.A. $y=2$

(ii) $\dfrac{x^2}{9}+\dfrac{y^2}{25}=1\Rightarrow\dfrac{x^2}{3^2}+\dfrac{y^2}{5^2}=1$

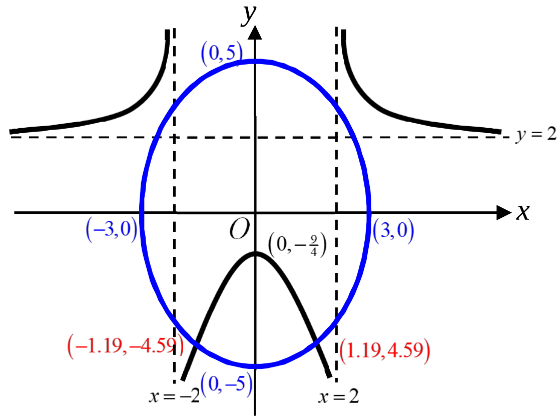
Ellipse, centre (0, 0).

'Width' ±3, 'height" ±5

$$\frac{y^2}{25}=1-\frac{x^2}{9}$$

$$y^2=25\left(1-\frac{x^2}{9}\right)$$

$$y=\pm5\sqrt{1-\frac{x^2}{9}}$$



[Draw proportionately, 'egg standing up"]

(iii)

Using GC, the intersections are $(-1.19,-4.59)$ and $(1.19,4.59)$

The x-coordinates are –1.19 and 1.19

(iv)

Observe that $y=-5\sqrt{1-\dfrac{1}{9}x^2}$ is the lower half of ellipse $\dfrac{x^2}{9}+\dfrac{y^2}{25}=1$

$$-5\sqrt{1-\frac{1}{9}x^2}\le\frac{2x^2+9}{x^2-4}$$

So $-3\le x<-2$ or $-1.19\le x\le1.19$ or $2<x\le3$

11. Sigmoid functions are used to model many natural processes such as population growth of virus. One example of a Sigmoid function f is given by

$$\mathrm{f}: x \mapsto \frac{1}{1+\mathrm{e}^{-x}}, \quad x \in \mathbb{R}$$

(i) Sketch the graph of $y = \mathrm{f}(x)$, indicating clearly the equation(s) of any asymptote(s) and the coordinates of any points where the curve crosses the axes. [2]

(ii) Find $\mathrm{f}^{-1}(x)$ in similar form. [3]

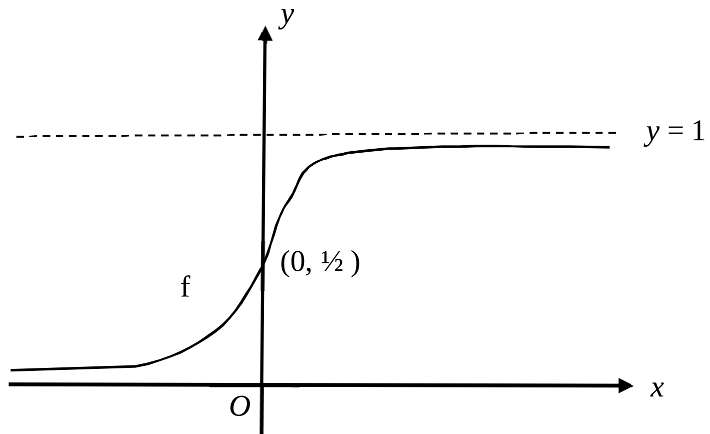
Another function g is given by $\mathrm{g}: x \mapsto 3x-1, \; x \in \mathbb{R}, \; 0 \le x \le 2$.

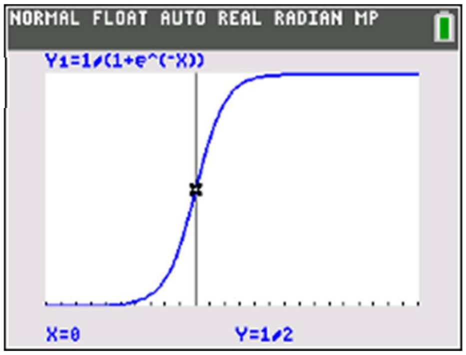
(iii) Show that fg exists and find the range of fg, expressing your answer in terms of e. [4]

(iv) Describe a sequence of transformations which transform the graph of y = f(x) onto the graph of y = fg(x). [2]

Solution

(i)





y-intercept at (0, ½)
HA: y = 1 and y = 0

(ii)

Let $y = \dfrac{1}{1+e^{-x}}$

$e^{-x} = \dfrac{1}{y} - 1$

Taking ln on both sides,

$-x = \ln\left(\dfrac{1}{y} - 1\right)$

$x = -\ln\left(\dfrac{1}{y} - 1\right)$

$\mathrm{f}^{-1}: x \to -\ln\left(\dfrac{1}{x} - 1\right), x \in (0,1)$ [From graph, R_f is $0 < y < 1$]

(iii)

Given $g : x \mapsto 3x-1,\ \ x \in \mathbb{R},\ \ 0 \le x \le 2$

[Since g is a straight line, to find R_g, just find the end-points $g(0) = -1$ and $g(2) = 5$]

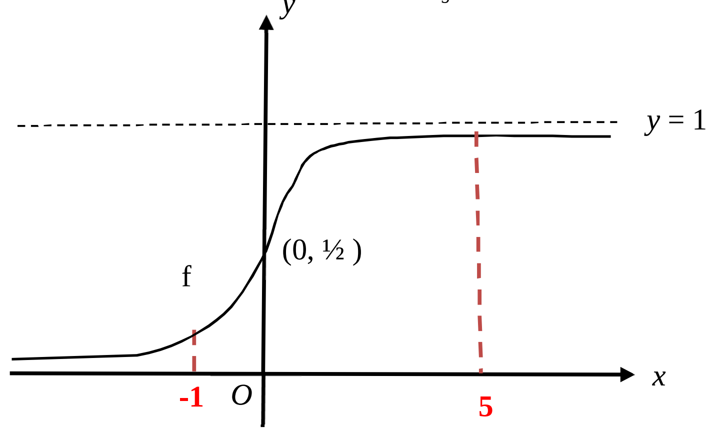
$R_g = [-1, 5]$

This is clearly a subset of $\mathbb{R} = D_f$

Alternatively,

If student can state that since $D_f = \mathbb{R}$,

so definitely $R_g \subseteq D_f$ so composite function fg exists. (i.e. do not need to find R_g.)



$D_{fg} = D_g = [0,2]$.

$fg(0) = f(-1) = \dfrac{1}{1+e}$

and $fg(2) = f(5) = \dfrac{1}{1+e^{-5}} = \dfrac{e^5}{e^5+1}$

Thus $R_{fg} = [\dfrac{1}{e+1}, \dfrac{1}{1+e^{-5}}]$.

(iv)

$fg(x) = f(3x-1)$

$f(x)$

$\downarrow$ Translate by 1 unit in the positive *x*-axis direction

$f(x-1)$

$\downarrow$ Scale parallel to the *x*-axis with factor $\dfrac{1}{3}$

$f(3x-1)$

Alternatively,

$f(x)$

$\downarrow$ Scale parallel to the x-axis with scale factor $\dfrac{1}{3}$

$f(3x)$

$\downarrow$ followed by a translation of $\dfrac{1}{3}$ units in the positive x-axis direction.

$f\left(3\left[x-\dfrac{1}{3}\right]\right) = f(3x-1)$

12. **(i)** Amanda wants to make a mask holder. To do so, she cuts out 4 identical equilateral triangles of sides 1 cm and a rectangular strip $\frac{x-1}{16}$ cm by $\frac{2y-1}{4}$ cm from a rectangle of sides x cm by y cm (see Diagram A). The rectangular strip has an area of 5.5 cm^2.

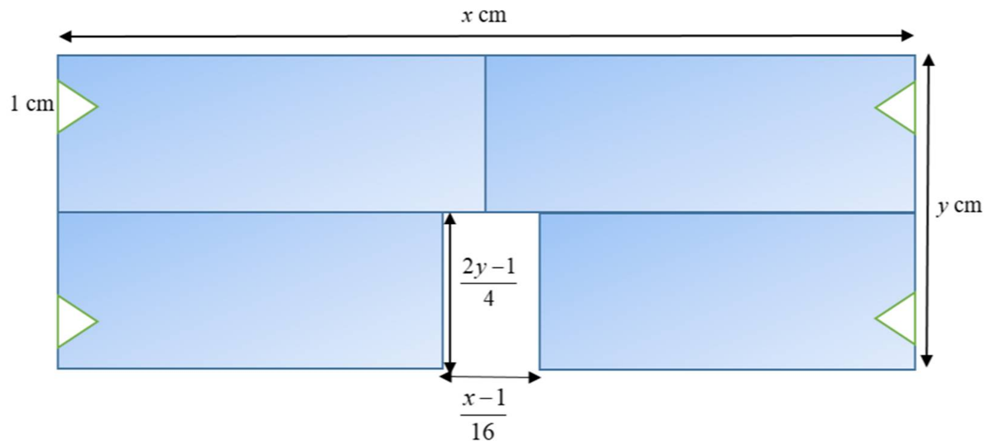


Diagram A

Show that A, the area of the mask holder is given by

$$A = \frac{176x}{x-1} + \frac{x}{2} - \sqrt{3} - \frac{11}{2}$$

Using differentiation, find the exact value of x such that area of the mask holder is a minimum. Hence determine the minimum area of the mask holder. [9]

(ii) The area of the rectangular strip is kept at 5.5 cm^2. Beth suggests that if the 4 identical triangle cut-outs are isosceles instead, the area of the mask holder could be made smaller as compared to the area found in part **(i)(b).**

For each of the triangles, let θ be the angle between two edges of length 1 cm each (see diagram B). Determine if Beth's suggestion is correct.

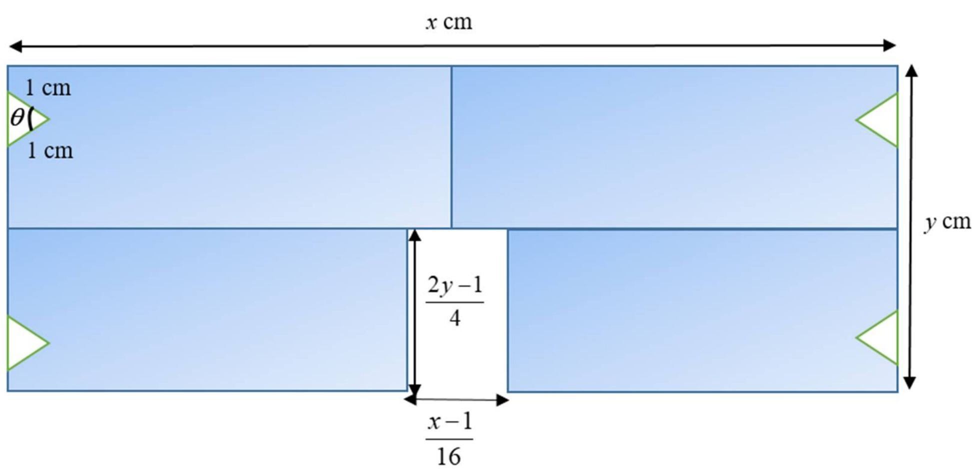


Diagram B

[2]

Solution

(i)
Area of mask holder:

$$A = xy - 4\left(\frac{1}{2}\right)(1)(1)\sin 60 - 5.5$$

$$A = xy - 4\left(\frac{1}{2}\right)\frac{\sqrt{3}}{2} - 5.5 \quad \text{-------------------- (1)}$$

Area of rectangular strip:

$$5.5 = \frac{2y-1}{4}\left(\frac{x-1}{16}\right)$$

$$11(2)(16) = (x-1)(2y-1)$$

$$y = \frac{1}{2}\left[\frac{11(2)(16)}{x-1} + 1\right] \text{----------------------- (2)}$$

Subst (2) into (1):

$$A = x\left\{\frac{1}{2}\left[\frac{11(2)(16)}{x-1} + 1\right]\right\} - \sqrt{3} - 5.5$$

$$A = \frac{176x}{x-1} + \frac{x}{2} - \sqrt{3} - \frac{11}{2} \quad \text{(shown)}$$

$$\frac{dA}{dx} = 176x(-1)(x-1)^{-2} + (x-1)^{-1}(176) + \frac{1}{2}$$

$$\frac{dA}{dx} = \frac{-176x + 176(x-1) + \frac{1}{2}(x-1)^2}{(x-1)^2}$$

$$0 = -176 + \frac{1}{2}(x-1)^2$$

$$11(16) = \frac{1}{2}(x-1)^2$$

$$11(16)(2) = (x-1)^2$$

$$x - 1 = \pm\sqrt{11(16)(2)} = \pm 4\sqrt{22}$$

$x = 4\sqrt{22} + 1$ (since $x > 0$, reject $x = -4\sqrt{22} + 1$)

To determine if $x = 4\sqrt{22} + 1$ gives a minimum area:

$$\frac{dA}{dx} = -11(16)(x-1)^{-2} + \frac{1}{2}$$

$$\frac{d^2A}{dx^2} = -11(16)(-2)(x-1)^{-3}$$

When $x = 4\sqrt{22} + 1$, $\frac{d^2A}{dx^2} = 11(16)(2)\left(4\sqrt{22}\right)^{-3} > 0$, thus $x = 4\sqrt{22} + 1$ gives a minimum value.

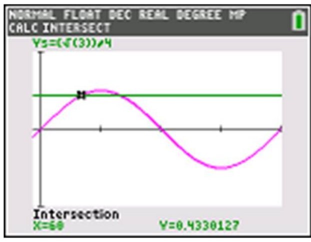
When $x = 4\sqrt{22}+1$, $y = \frac{1}{2}\left[\frac{11(2)(16)}{4\sqrt{22}}+1\right]$

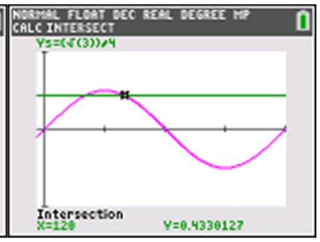
Thus, area $A = xy - 4\left(\frac{1}{2}\right)\frac{\sqrt{3}}{2} - 5.5 = \left(4\sqrt{22}+1\right)\left(\frac{1}{2}\left[\frac{11(2)(16)}{4\sqrt{22}-1}+1\right]\right) - 4\left(\frac{1}{2}\right)\frac{\sqrt{3}}{2} - 5.5 = 188 \text{ cm}^2$

(ii)

Area of 1 triangle = $A = \frac{1}{2}(1)(1)\sin\theta$

And when $\theta = 60°$, $A = \frac{1}{2}(1)(1)\sin 60° = \frac{\sqrt{3}}{4}$





So for any value $60° < \theta < 120°$, it will give a bigger value of A than when it is an equilateral triangle.

And so, the area of the face mask holder will be a minimum (with the area of the rectangular strip remaining at 5.5 cm^2) if the areas of the 4 triangles cut-outs are a maximum,
thus Beth's suggestion is correct.
